

Regression and hypothesis testing with a categorical explanatory variable

Warmup

Work on the warmup activity (handout), then we will discuss as a class.

Warmup

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.80642	1.93822	0.932	0.3520
Depth	0.06889	0.03001	2.296	0.0223 *

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Depth	1	43.66	43.657	5.2711	0.02227 *
Residuals	349	2890.53	8.282		

How do the p-values and test statistics relate?

t-tests and F-tests

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For the simple linear regression model

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

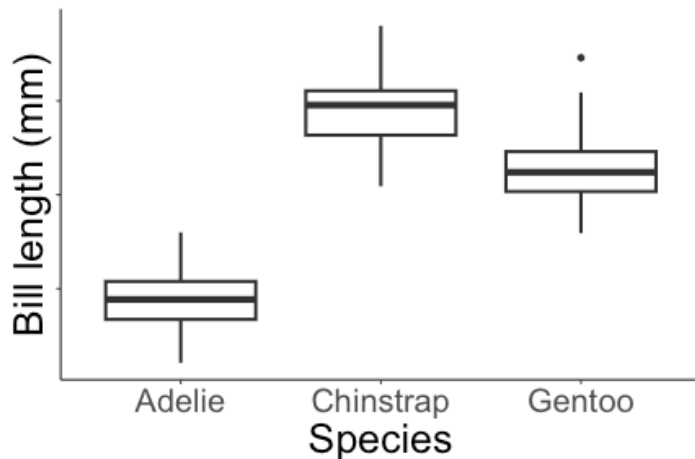
consider testing $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$.

✚ $t^2 = F$

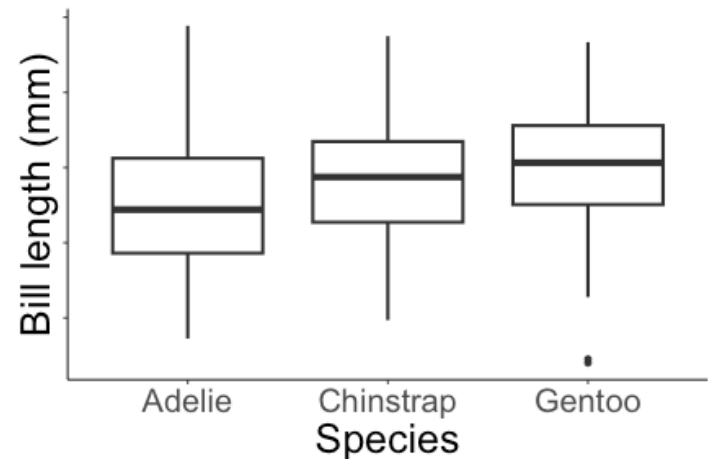
✚ p-values are the same for a t-test and an F-test

Previously

More evidence for a relationship when *between-group* variability is larger than *within-group* variability.



Large between-group variability, relative to within-group variability.



Small between-group variability, relative to within-group variability.

Analysis of variance

Test statistic:

$$F = \frac{\frac{1}{p-1} \sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\frac{1}{n-p} \sum_{i=1}^n (y_i - \hat{y}_i)^2} = \frac{\frac{1}{p-1} SS_{Model}}{\frac{1}{n-p} SSE} = \frac{MS_{Model}}{MSE}$$

Analysis of variance (ANOVA) table:

Source	df	SS	MS	F
Model	$p - 1$	SS_{Model}	$MS_{Model} = \frac{SS_{Model}}{p-1}$	$F = \frac{MS_{Model}}{MSE}$
Residual	$n - p$	SSE	$MSE = \frac{SSE}{n-p}$	
Total	$n - 1$	SST_{Total}		

Last time: diamonds

Data on 351 diamonds. Model:

$$\text{Price per carat} = \beta_0 + \beta_1 \text{IsSI1} + \beta_2 \text{IsSI2} + \beta_3 \text{IsSI3} + \beta_4 \text{IsVS1} \\ + \beta_5 \text{IsVS2} + \beta_6 \text{IsVVS1} + \beta_7 \text{IsVVS2} + \varepsilon$$

$p = 8$

Source	df	SS	MS	F
Model	7	108.43	15.49	1.88
Residual	343	2826.32	8.24	
Total	350	2934.75		

$$\frac{MS_{\text{Model}}}{MSE} = \frac{15.49}{8.24}$$

$$n - p = 351 - 8$$

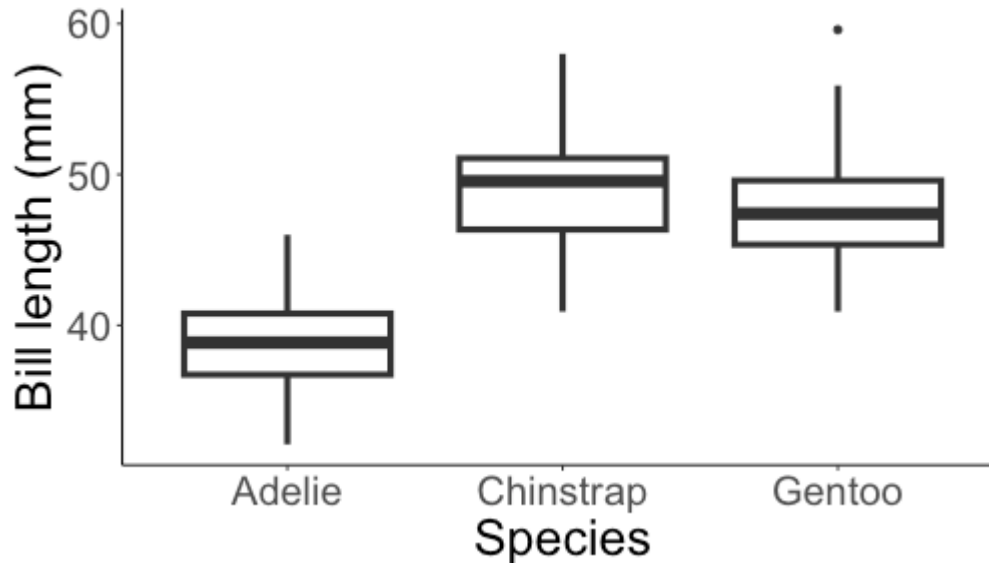
$$n - 1$$

$$SS_{\text{Model}} = MS_{\text{Model}} \times (p - 1) = 15.49 \times 7 = 108.43$$

$$SS_E = MSE \times (n - p) = 8.24 \times 343 = 2826.32$$

$$SS_{\text{Total}} = SS_E + SS_{\text{Model}}$$

Test statistic



Test statistic: $F = \frac{\text{between group variance}}{\text{within group variance}} = 397.3$

Are large or small values of F evidence against H_0 ?

large values are evidence against H_0 (no relationship)

if $t \sim t_d$ then $t^2 \sim F_{1,d}$

F distribution

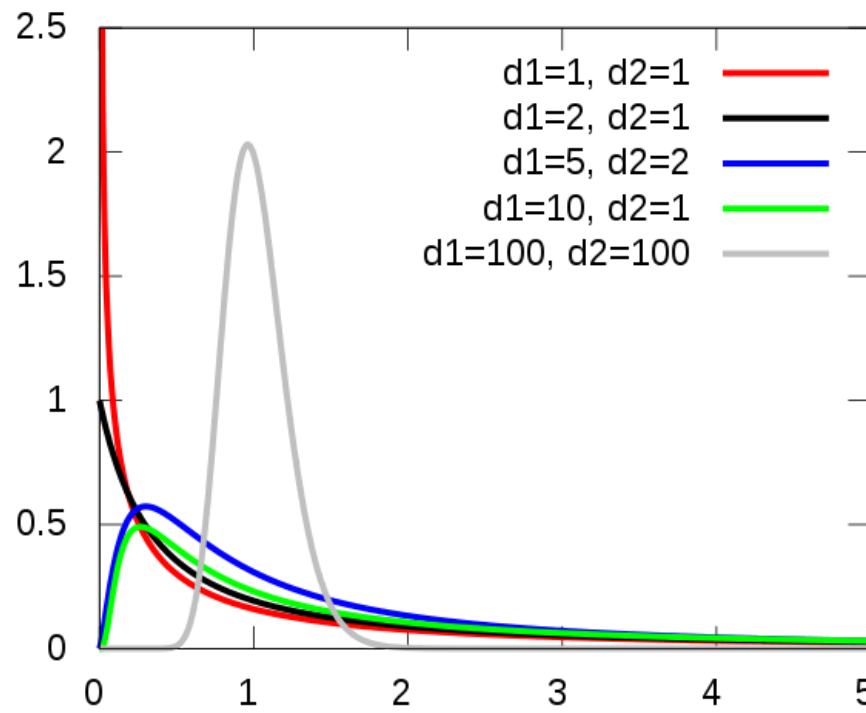
If H_0 is true, then $F = \frac{MSModel}{MSE}$ follows an $F_{p-1, n-p}$ distribution.

numerator df
denominator df

$F \geq 0$ b/c $SSModel \geq 0$

Some F_{d_1, d_2} distributions:

$SSE \geq 0$

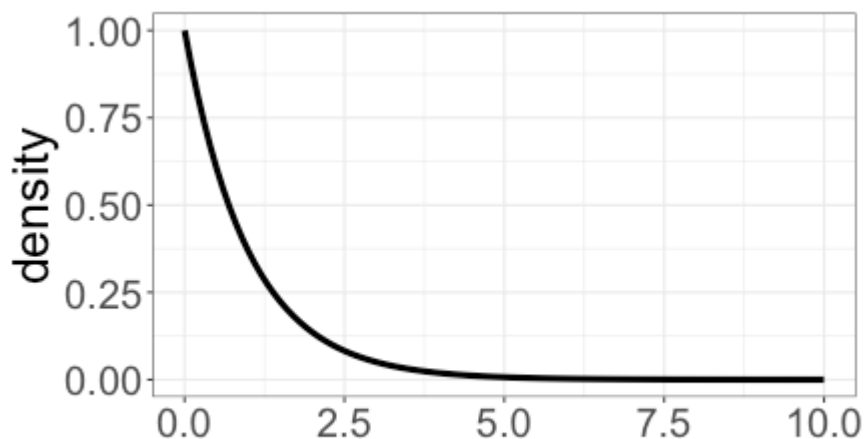


Calculating p-values

Test statistic: $F = 397.3$

Degrees of freedom: $p - 1 = 2, n - p = 330$

$F_{2, 330}$ distribution:



p-value = $P(F_{2, 330} > 397.3) \approx 0$

Summary

Model: bill length = $\beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$

Hypotheses:

$$H_0 : \beta_1 = \beta_2 = 0$$

$$H_A : \text{at least one of } \beta_1, \beta_2 \neq 0$$

Test statistic: $F = 397.3$

p-value ≈ 0

Conclusion: We have strong evidence that there is a relationship between species and bill length.

In R

linear model object
i.e. species_lm <- lm(...)

anova(species_lm)

```
## Analysis of Variance Table
```

```
##
```

```
## Response: bill_length_mm
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
```

```
## species      2 7015.4   3507.7    397.3 < 2.2e-16 ***
```

```
## Residuals 330 2913.5         8.8
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
```

Source	df	SS	MS	F
Model	2	7015.4	3507.7	397.3
Residual	330	2913.5	8.8	
Total	332	9928.9		

Class activity

- + For the remainder of class, work on the class activity
 - + Regression, diagnostics, and inference with the diamonds data
- + Submit rendered HTML on Canvas when finished